

Dibbayajyoti Roy

Full Stack Engineer • TypeScript • React • PostgreSQL • Rust

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SUMMARY

Full-stack engineer with one year of full-time production experience shipping TypeScript/React frontends, Node.js services, and PostgreSQL-backed systems end to end, from schema design through deployment on Linux infrastructure I administer myself. Additionally, six merged pull requests in two production Rust codebases (Meta's pyrefly, ReductStore), evidence of code quality that survives strict maintainer review. Comfortable owning features under direct customer pressure. Willing to relocate to Warsaw and subsequently to the US.

PROFESSIONAL EXPERIENCE

Full-Stack Software Engineer, Yupcha Softwares Pvt. Ltd. • Jul 2025 to Present • Agartala, India

- Shipped HR SaaS modules used across partner organisations (records, leave workflow, role-based access): 12+ Node.js REST endpoints over PostgreSQL with schema design, Row-Level Security, and composite indexing; deployed to Ubuntu VMs on Proxmox with systemd, nginx reverse proxies, and automated TLS renewal.
- Cut page load from 3.4s to 1.9s (~44%) by eliminating N+1 queries, adding composite indexes, and route-splitting Next.js bundles; containerised services with Docker.
- Rebuilt a production speech-to-speech AI interviewer end to end from a high-latency inherited codebase: migrated a cascaded STT/LLM/TTS pipeline to unified speech-to-speech on AWS Bedrock (Nova Sonic) with Pipecat over LiveKit (WebRTC), then owned it in production.
- Cut a hot Redis polling path ~90% by rewriting SCAN+GET as MGET batching; diagnosed a multi-track audio/video sync defect down to per-track delay compensation in FFmpeg filter graphs (LiveKit egress), integration debugging across three vendor layers.

OPEN SOURCE: CODE QUALITY UNDER MAINTAINER REVIEW

Six merged pull requests in two production Rust codebases; every change landed through strict external code review with tests and documentation.

- **ReductStore** (time-series database, Rust): five merged PRs in the observability layer, including a maintainer-requested cross-cutting refactor unifying all system event handling (#1496) and replication diagnostics shipped in v1.20 with public credit from the co-founder (#1417).
- **pyrefly** (Meta's Python type checker, Rust): fixed a false-positive untyped-import diagnostic per PEP 561, correctly handling submodules (#3840).

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript (ES2023), SQL, Rust, Python
- **Frontend:** React, Next.js (App Router), TanStack Query, Tailwind CSS; author of RoyUI, a TypeScript-first React component library
- **Backend:** Node.js, Hono, Express, REST, Server-Sent Events, WebRTC, Zod, OAuth, JWT
- **Databases:** PostgreSQL (schema design, Row-Level Security, partitioning, indexing), Redis, DynamoDB, MongoDB
- **Infrastructure:** Linux, Docker, Proxmox VE, nginx (TLS, rate limiting), systemd, AWS (Bedrock, DynamoDB, CloudWatch), Cloudflare Workers
- **Systems:** async Rust (tokio): channels, async locks, task-local state, reentrancy-safe design

SELECTED PROJECTS

- **AHTML** (TypeScript): agent-readable HTML endpoint standard emitting WebMCP, OpenAPI 3.1, JSON-LD, and llms.txt from one source; npm suite with 8.7k+ downloads.
- **whatbroke** (TypeScript CLI): crash-context packager for AI debugging; MCP server, git-anchored suspect ranking, parsers for five test runners; on npm.

HONORS & EDUCATION

Winner, NITA Arjuna 2.0 National Hackathon (2025), 200+ teams • Winner, Technovate Project Exhibition (2025) • 1st Runner-Up, NITA-ISRO Space Hackathon (2024)

B.Tech, Computer Science, ICFAI University Tripura (completed 2026; conferral October 2026)

English (professional), Bengali (native), Hindi (fluent) • Willing to relocate to Warsaw, Poland and to the US as required